

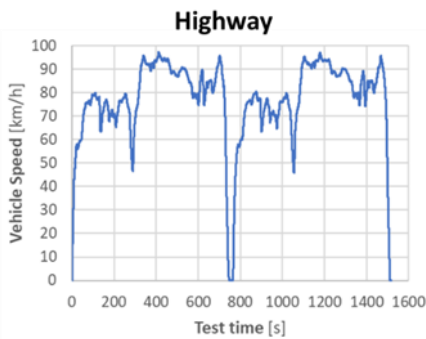
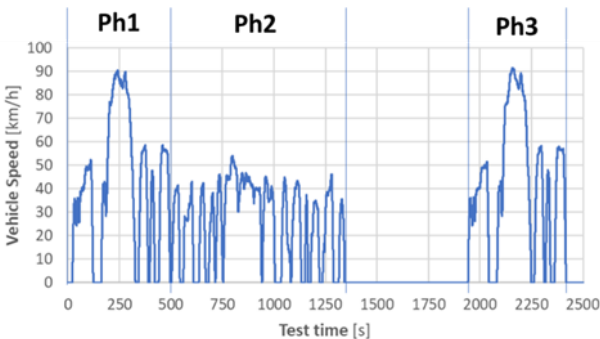
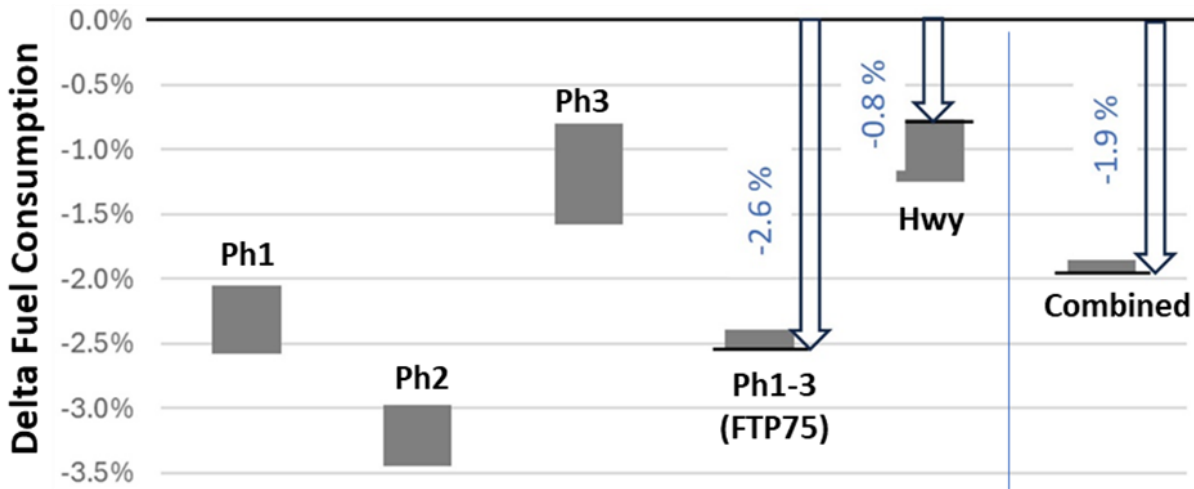
Lubrificantes

dezembro de 2024
P&D

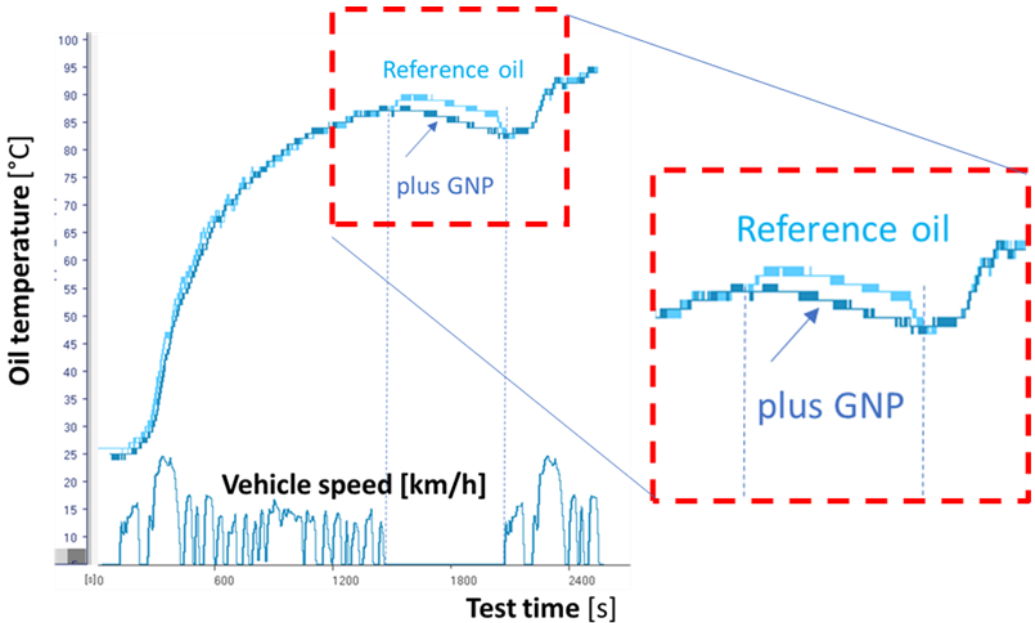


Tests at VW (with L66_2, 0.1%)

2.6% of fuel saving at Urban conditions, 0.8% at highway



Faster cooling. Higher thermal conductivity?



	Units	Ph1 & 3	Ph2	Highway
Duration	s	505	864	12.75
Distance	km	5.78	6.21	16.58
Mean Velocity	km/h	41.20	25.88	77.73
Max. Velocity	km/h	91.2	55.2	96.56
Stops		6	13	None

Paper submitted with VW and Idemitsu, and USP



Article

Effect of Graphene nanoplatelets as lubricant additive on fuel consumption during vehicle emission tests.

Eduardo Tomanik^{1*}, Wania Christinelli¹, Pamela Sierra Garcia¹, Scott Rajala², Jesuel Crepaldi³, Davi Franzosi⁴, Roberto Martins Souza⁴, Fernando Fusco Rovai^{5,6}

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- ⁵ VW do Brasil – Way to Zero Center; fernando.rovai@volkswagen.com.br
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Abstract: Lubricant friction modifier additives are used on lower-viscosity engine oils to mitigate the impact on the boundary friction. In this work, the use of Graphene nanoplatelets, with an average of 9 layers, was first investigated in a fully formulated SAE 0W-20 engine oil through the use of reciprocating tribological tests at 40 and 120 °C and Hertzian pressure up to 1.2 GPa. Then, engine fuel consumption was evaluated through fuel consumption in a vehicle emission test adding 0.1% of GNP, in an improved booster, on a fully formulated 5W-20 SAE oil. Adding 0.1% of GNP reduced fuel consumption by 2.6% in the FTP-75 cycle and 0.8% in the Highway.

CM018 – Desenvolvimento de aditivo químico base água contendo grafeno para ativação da escória em baixas idades para ganhos de resistência

Atualizações do desenvolvimento

dezembro de 2024
P&D



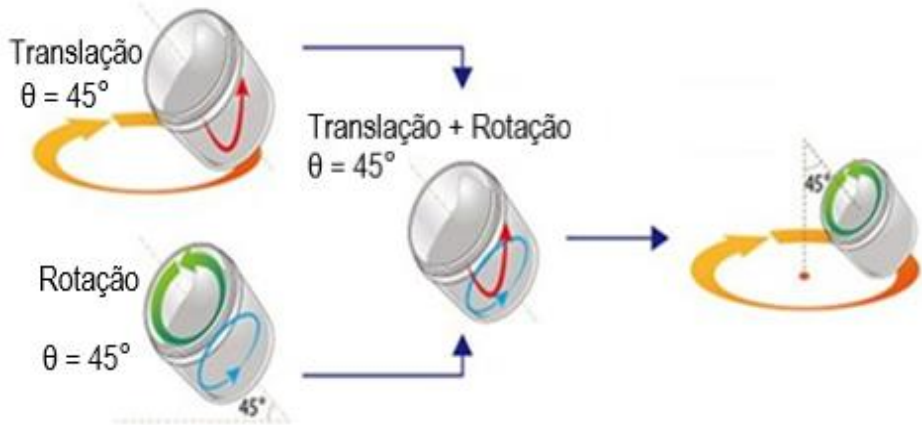
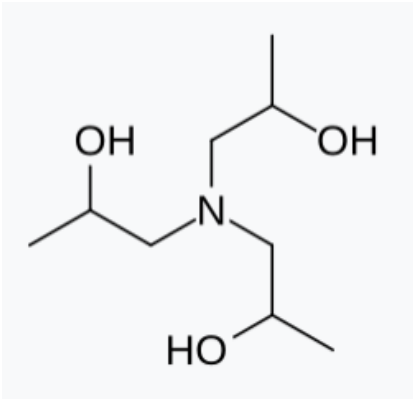


INFORMAÇÕES DO PROJETO	
Código & Nome do Projeto	CM018 – Aditivo Químico base água contendo Gr para ativação da escória em baixas idades para ganhos de resistência
Data de início de apropriação do ativo intangível	mai/24
Data do documento	mai/24
Plataforma	Construção
Segmento de Mercado	Construção Civil
Aplicações	Concretos estruturais com uso de adição mineral à base de escória de alto forno

PROCESSO DE PRODUÇÃO



+



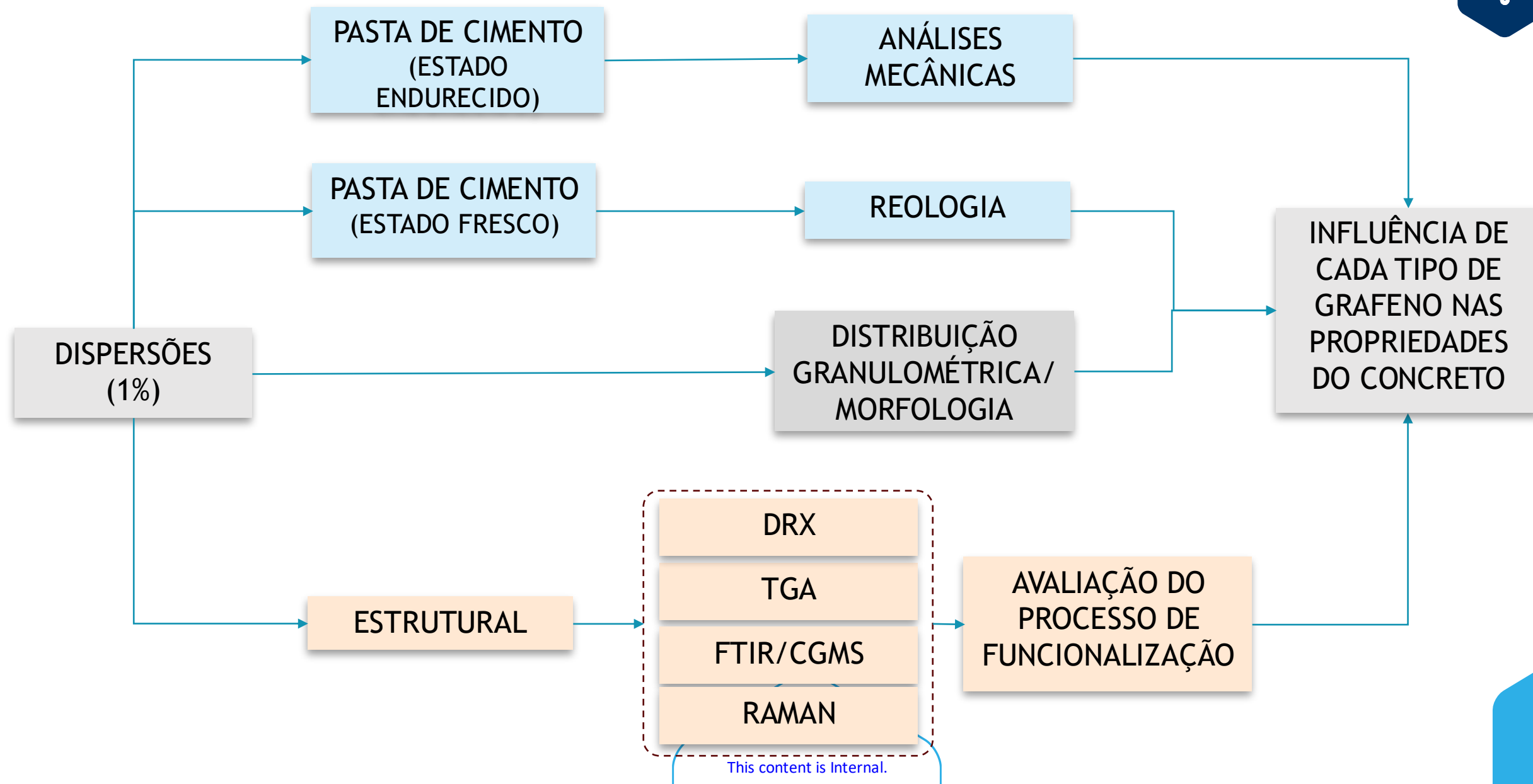
GRAFENOS AVALIADOS



0X	-	NanoXplore
BS3	-	BlackSwan
BS8	-	BlackSwan
FG05	-	First Graphene
FG10	-	First Graphene
FG20	-	First Graphene
HP10	-	Hexo
HP50	-	Hexo

A escolha inicial destes levou em conta a disponibilidade e morfologia

DESENVOLVIMENTO DE CONCENTRADOS DE GRAFENO

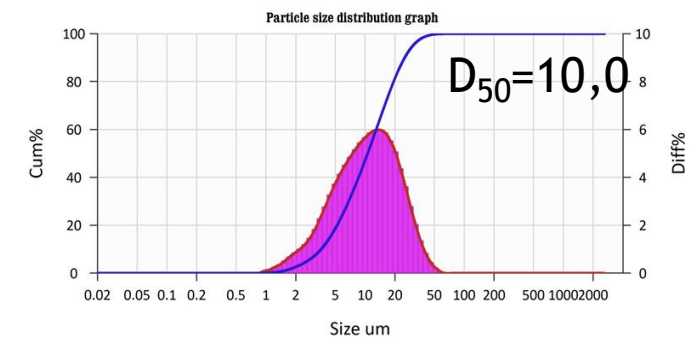
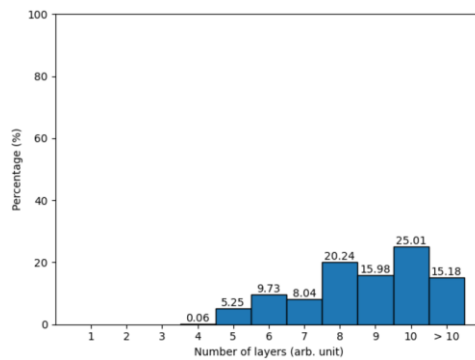


Concentrados

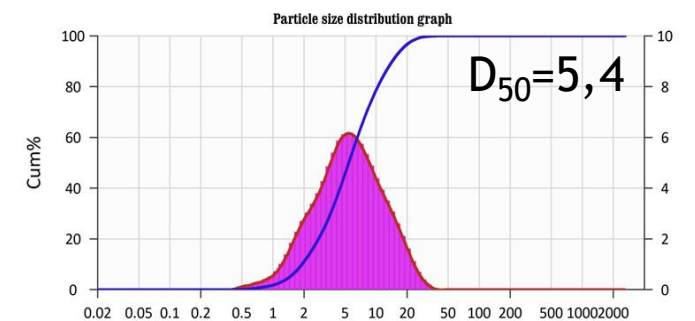
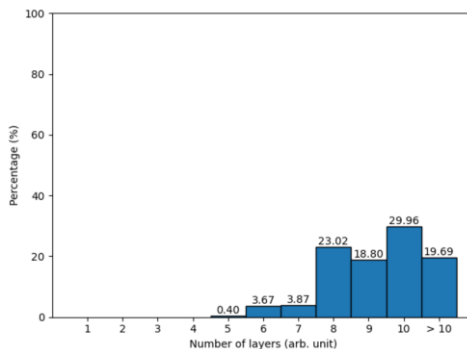


CM018

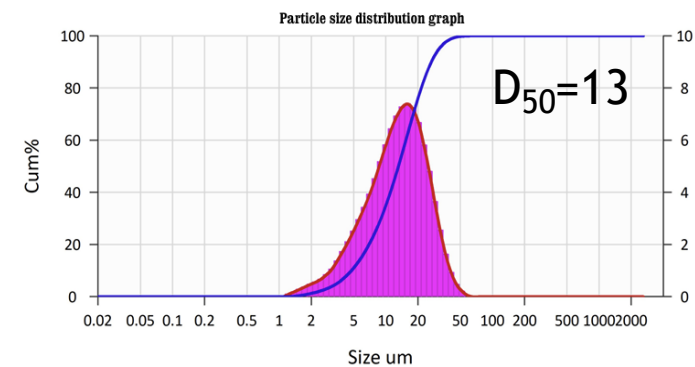
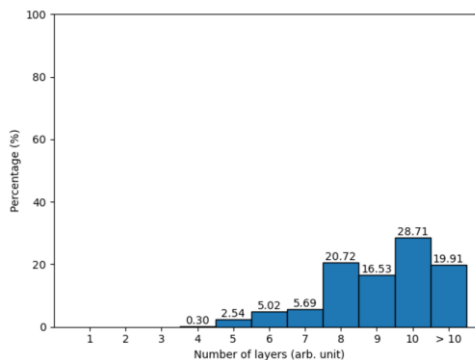
FG10



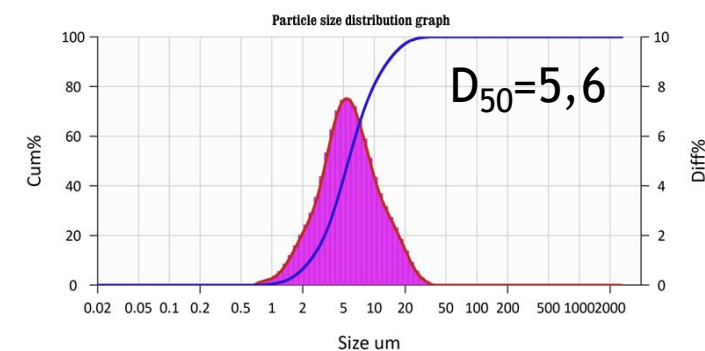
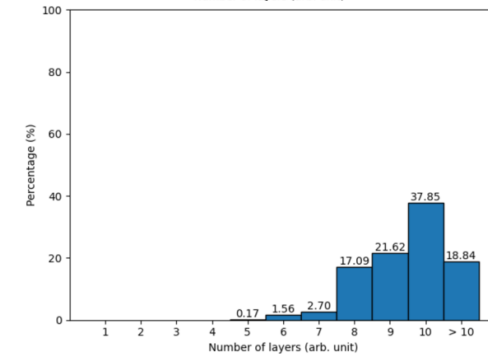
BS3



OX

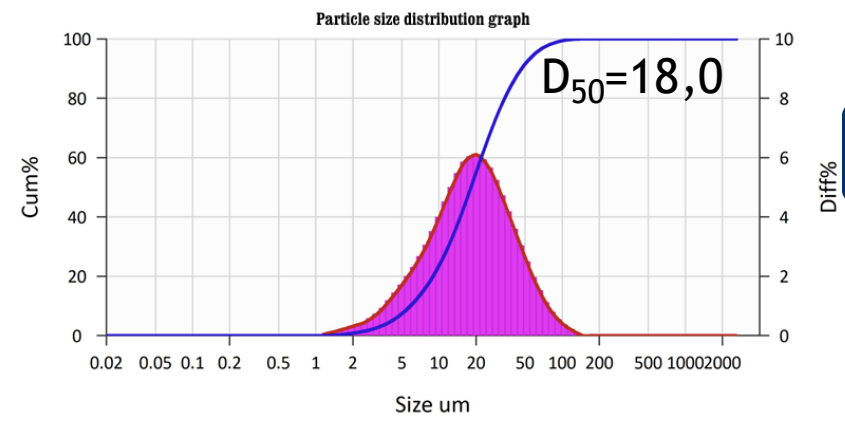
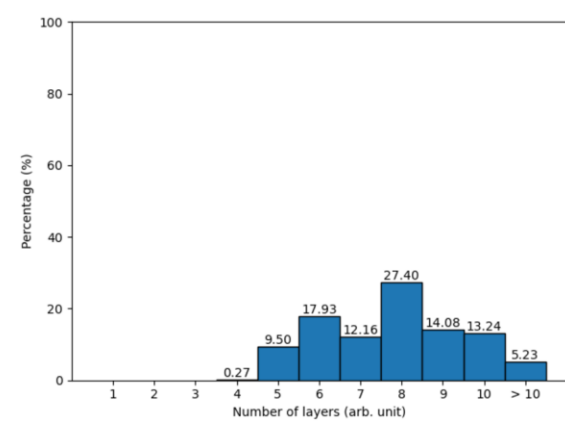


HP10

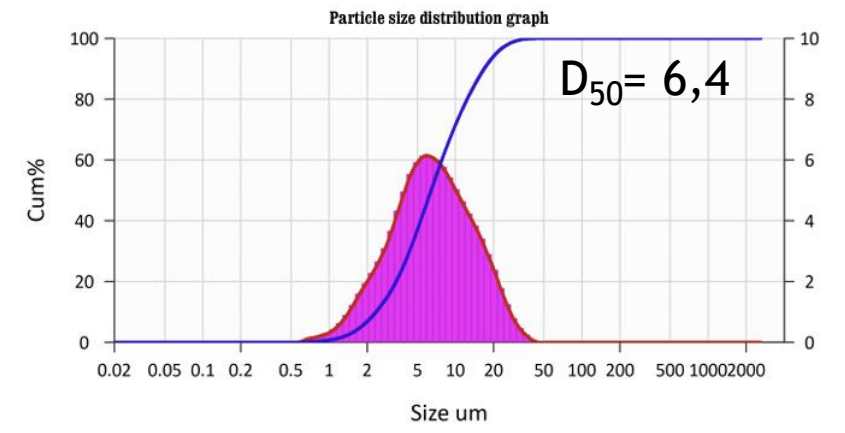
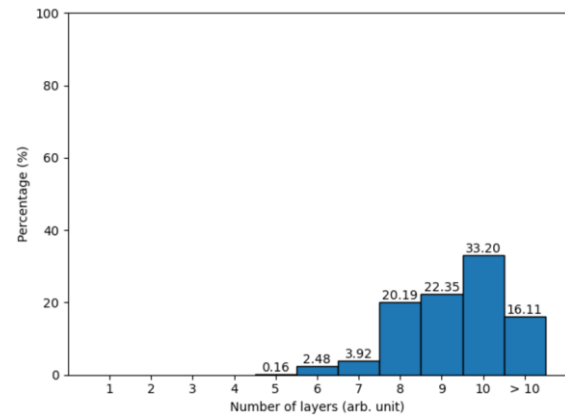


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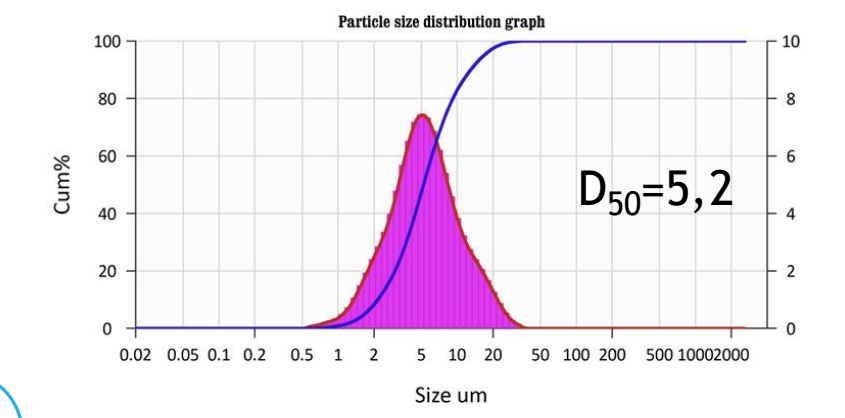
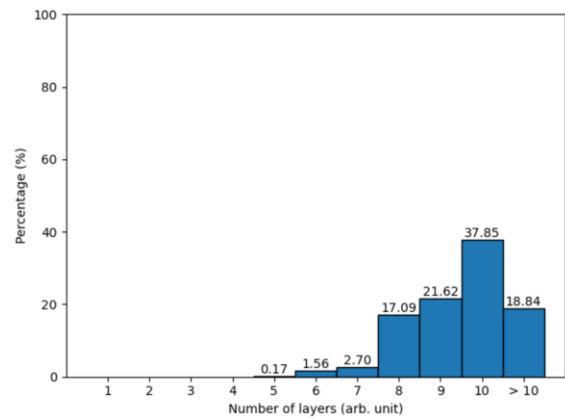
FG20



BS8



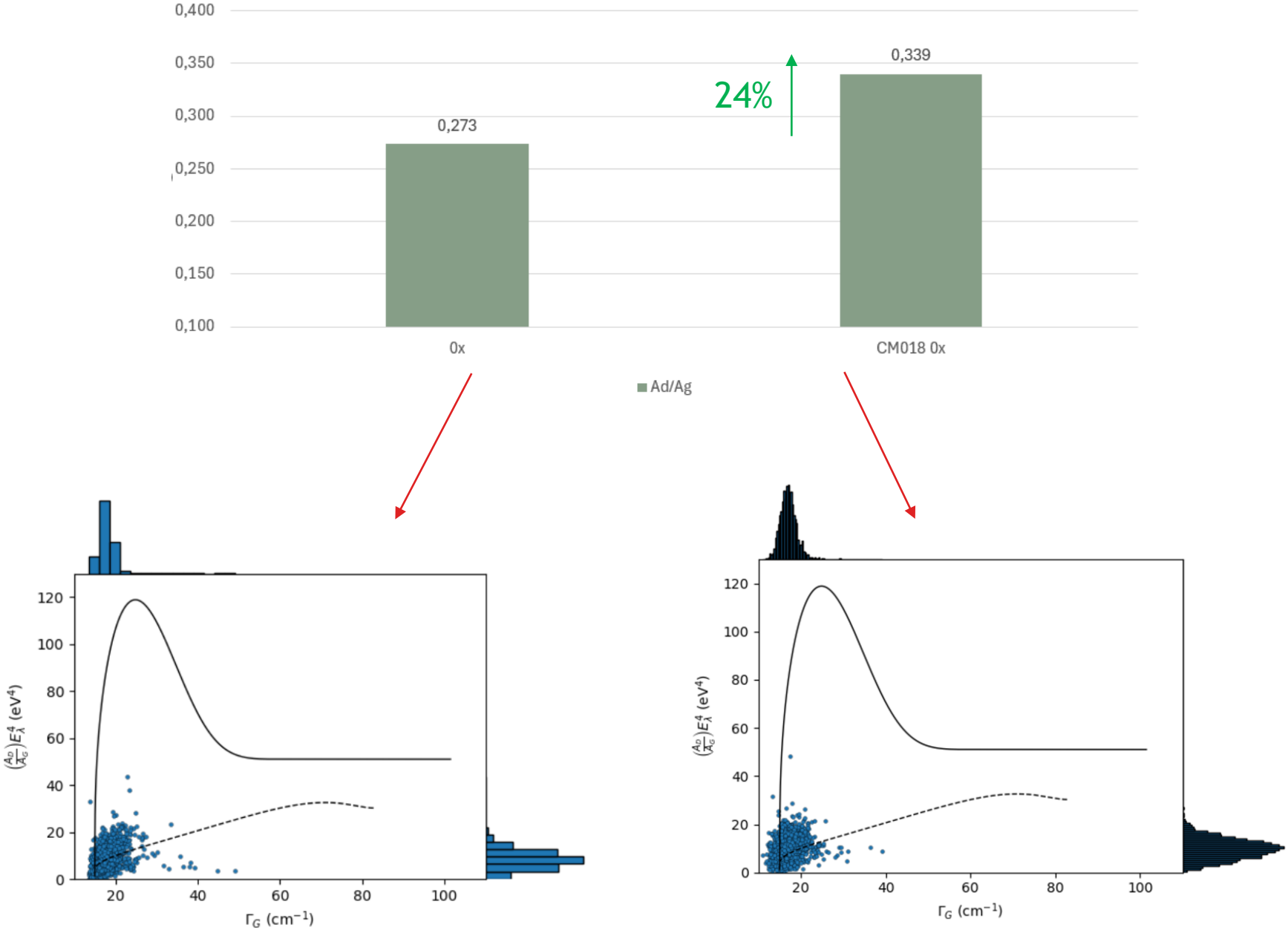
HP50



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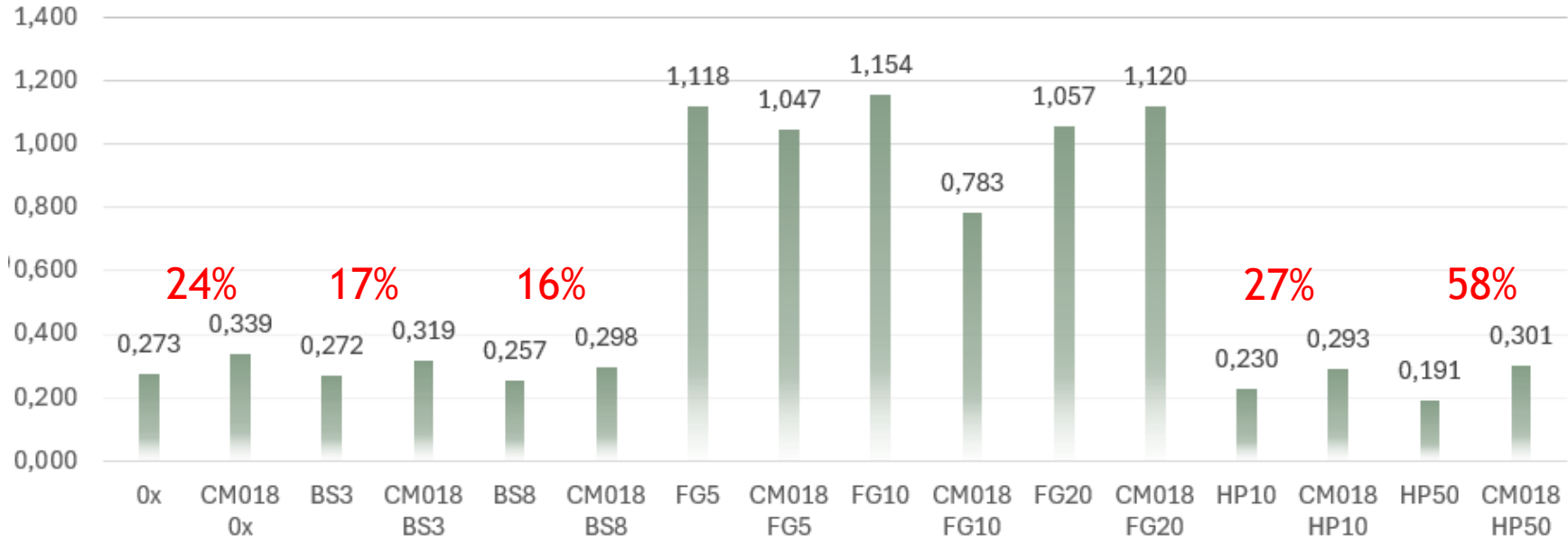
RAMAN

A_D/A_G

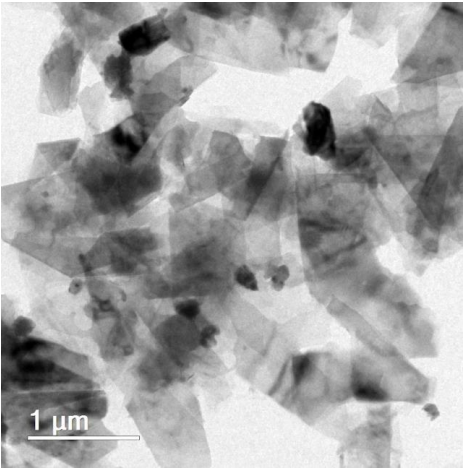


RAMAN

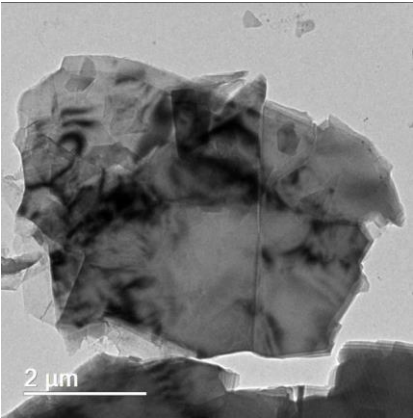
A_D/A_G



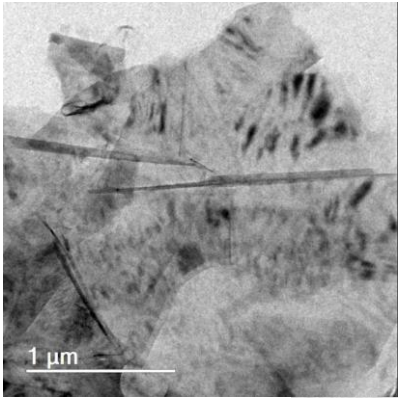
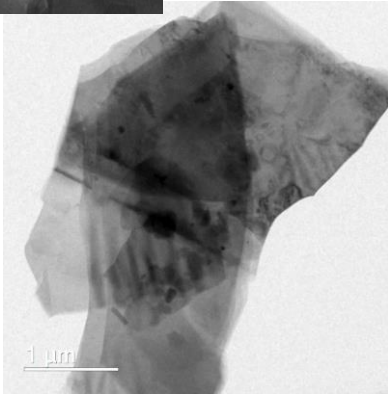
■ Ad/Ag



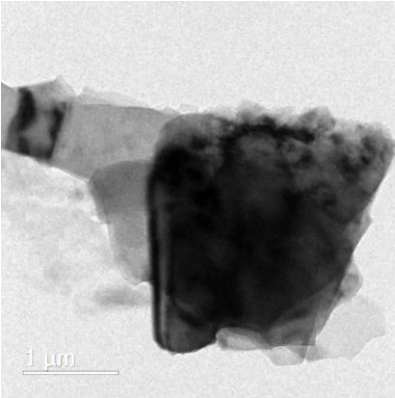
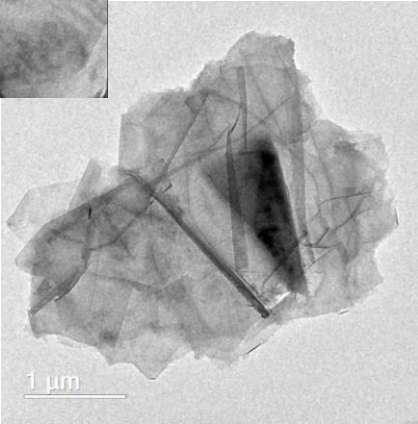
0x



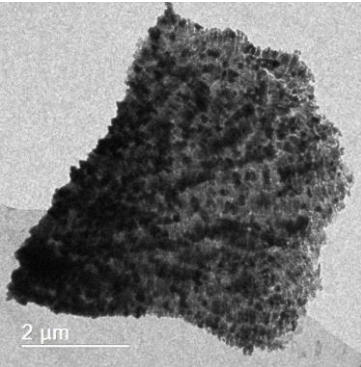
BS



FG



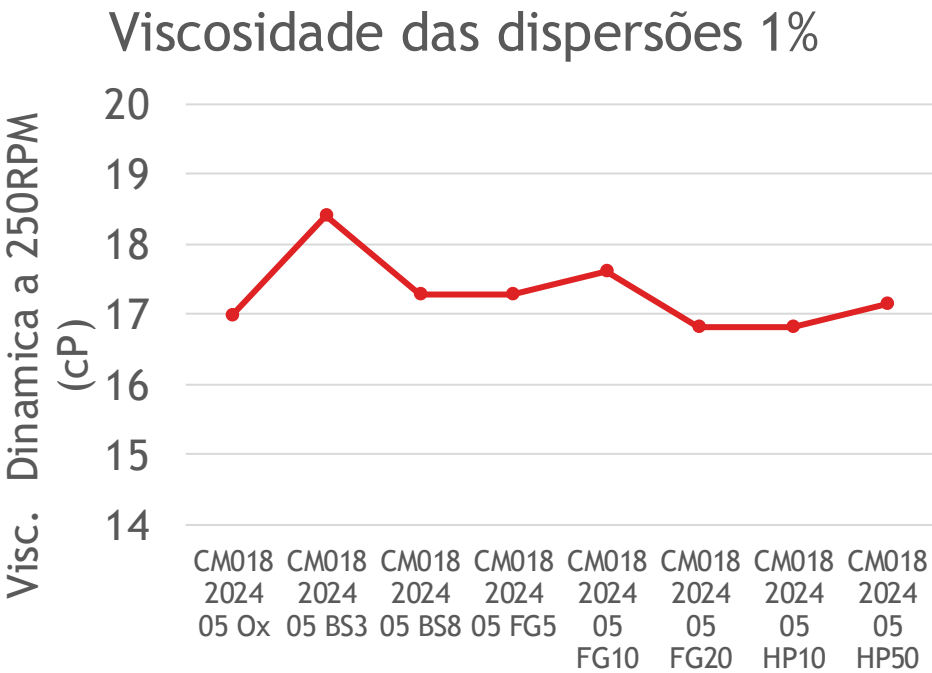
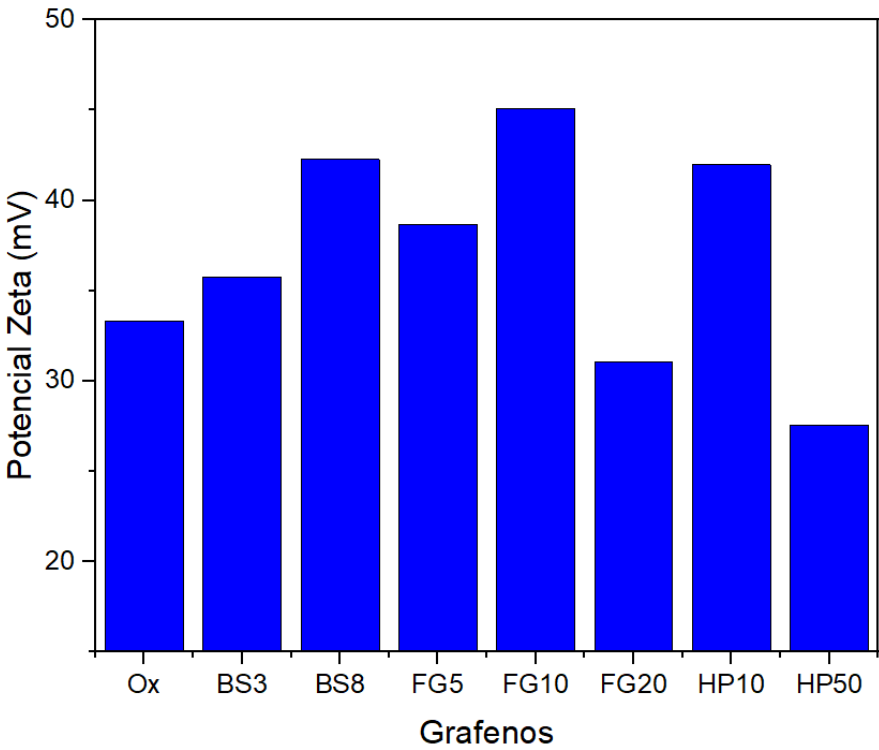
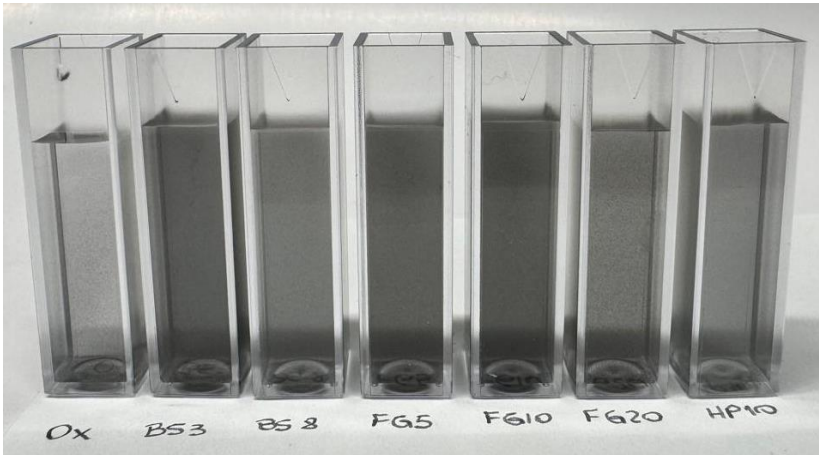
HP50



DISPERSÕES 1%

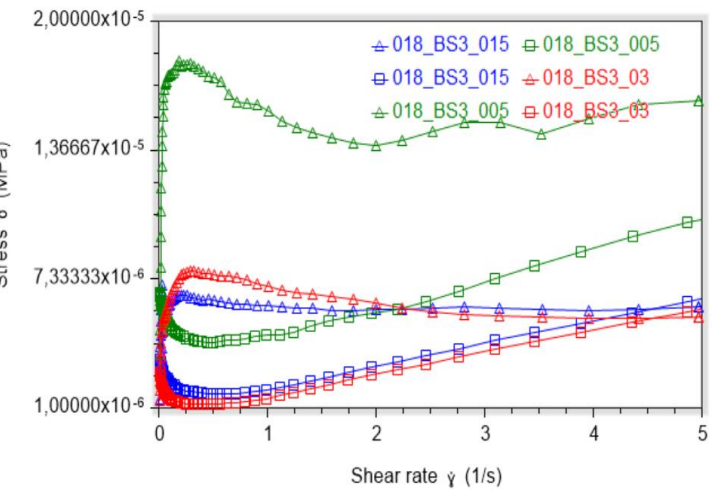


3000 RPM
5min

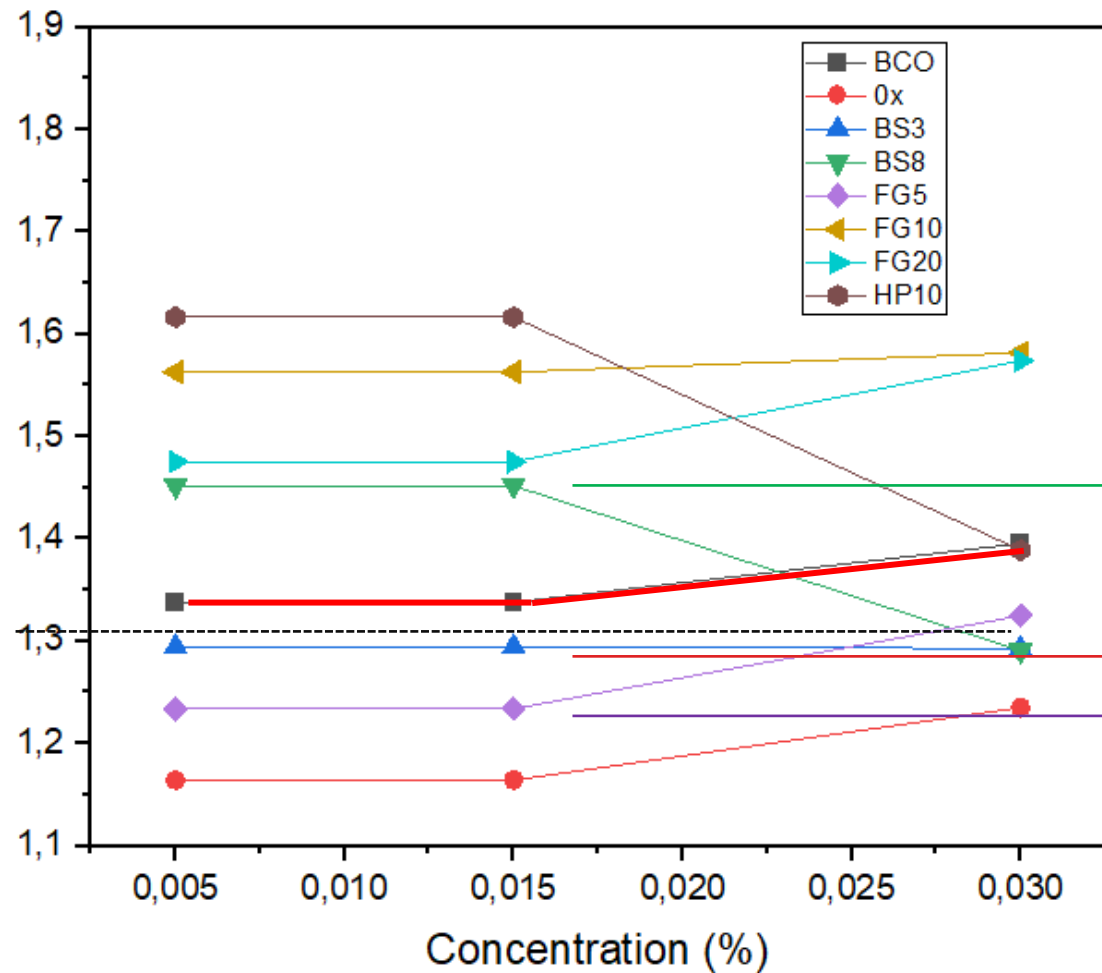


CARACTERIZAÇÕES

Flow curve



Viscosity (Pa.s)



BS8

BS3

FG5

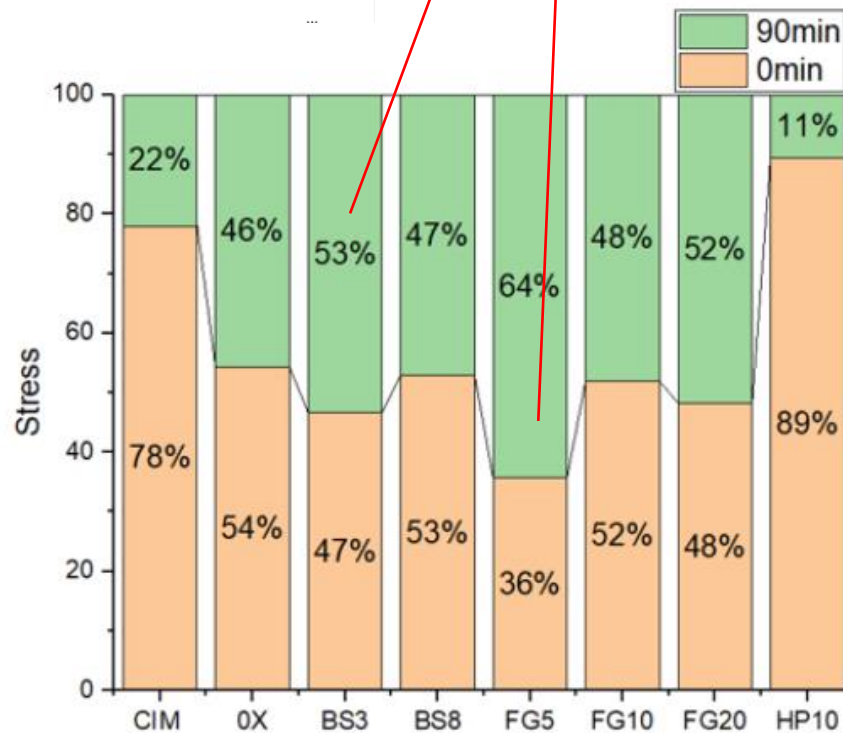
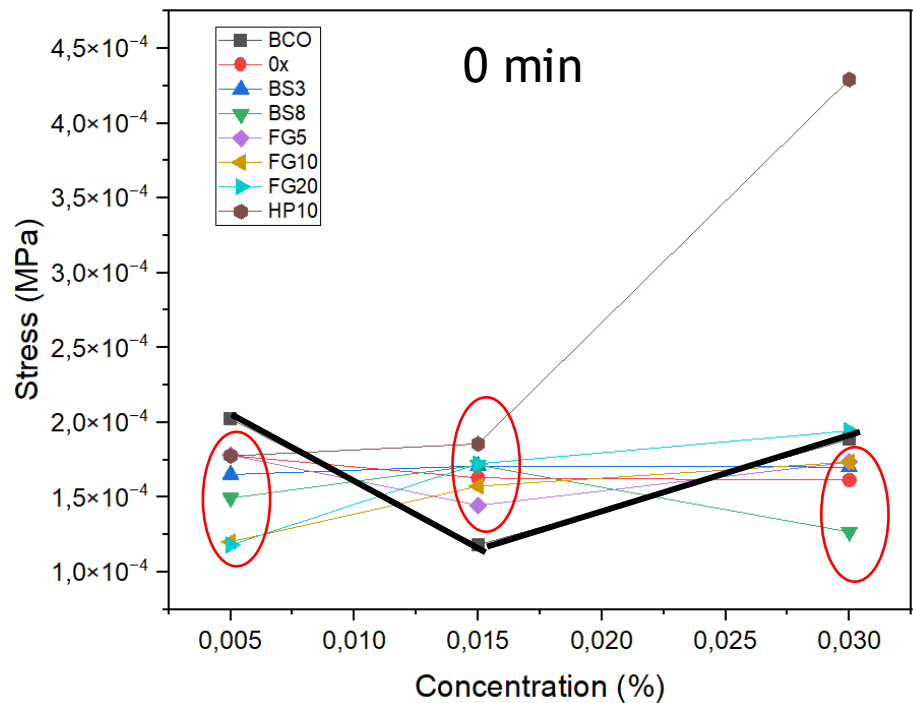
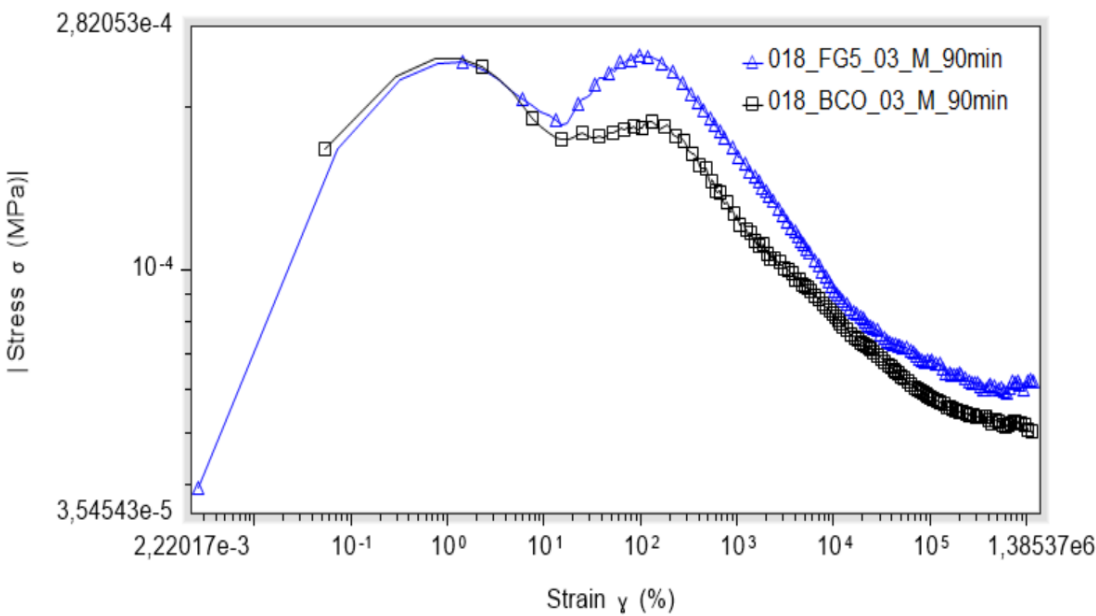
BS3 tem menor influência na reologia no estado fresco

Reologia em pasta de cimento

CARACTERIZAÇÕES

Stress Growth

Pasta de cimento



0,03%

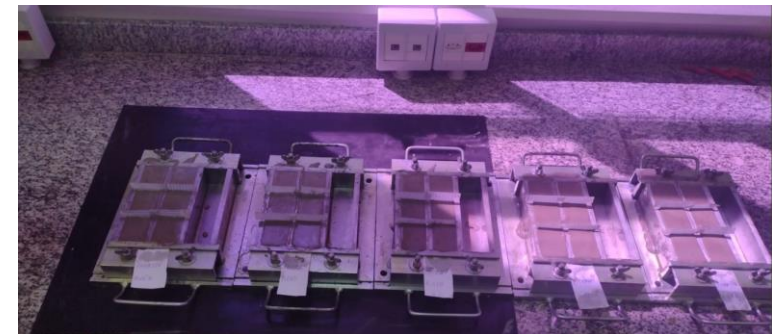
AVALIAÇÃO DE DESEMPENHO



Realizada em corpos de prova de pasta de cimento, sendo estes produzidos na forma prismática e com dimensões de 40 x 40 x 40 mm.

O programa de testes contou com a produção das pastas de cimento com teores de Grafeno de:

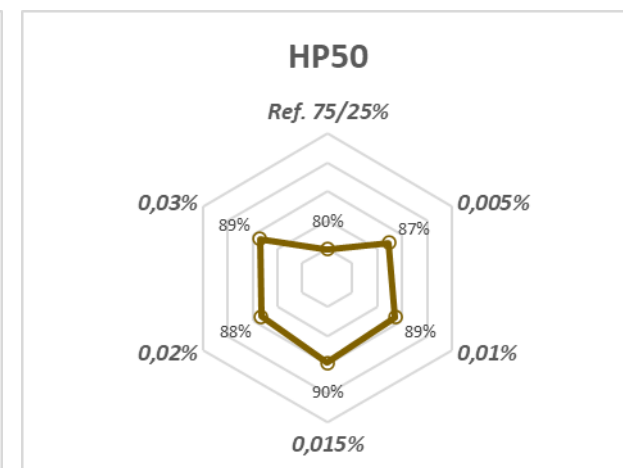
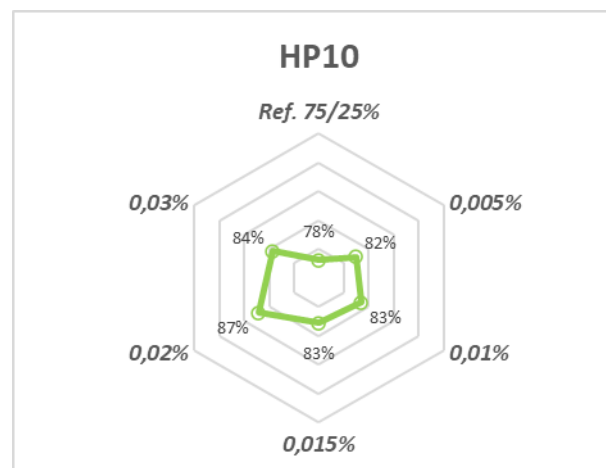
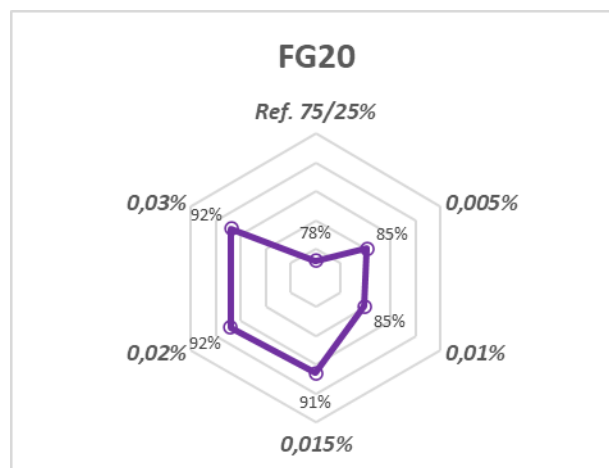
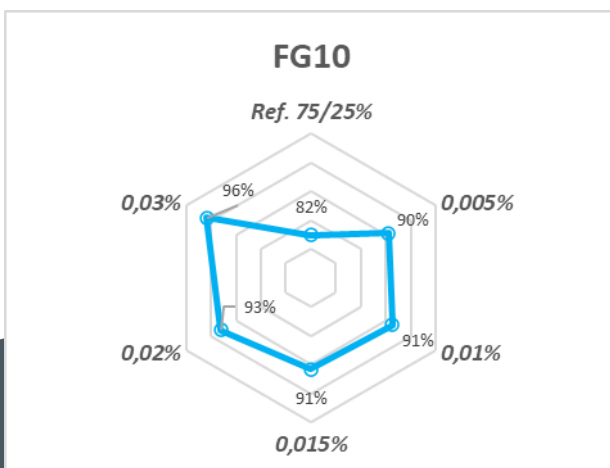
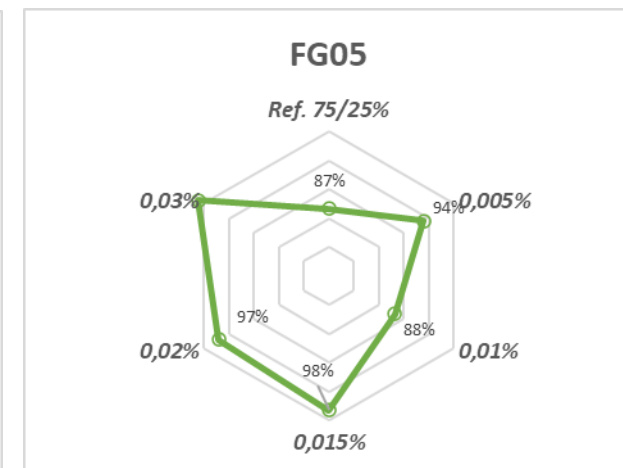
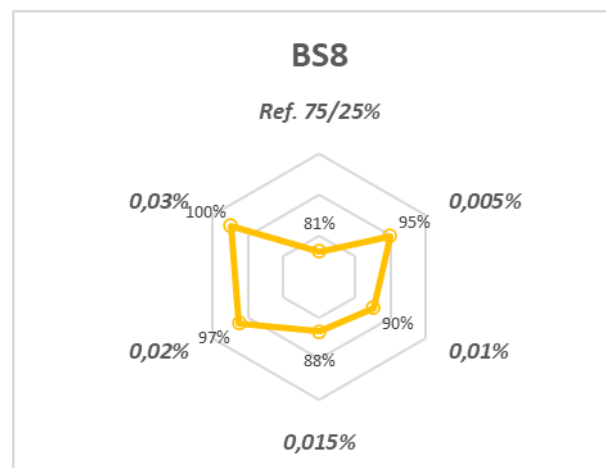
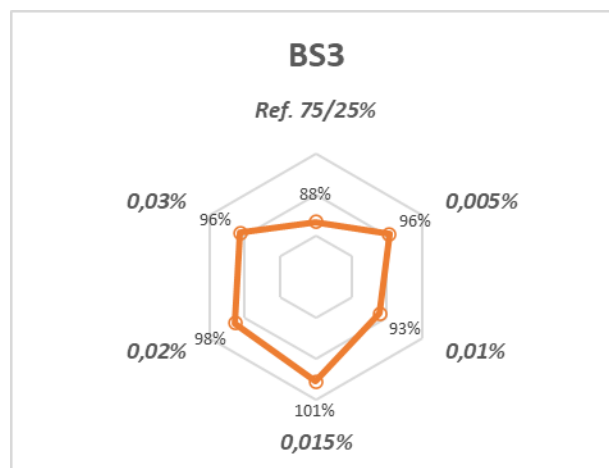
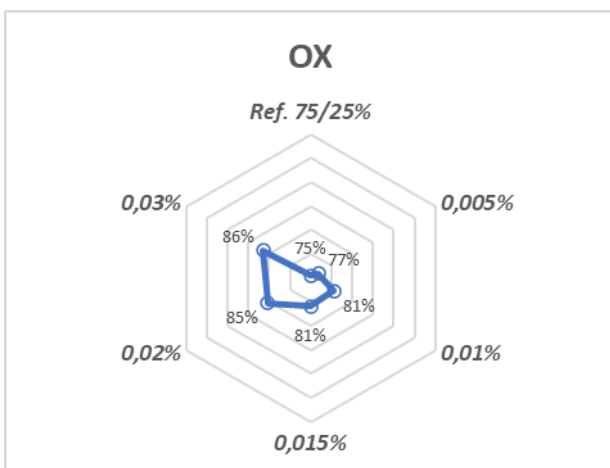
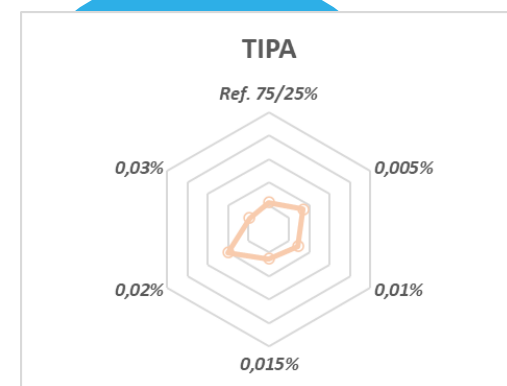
- 0,005%
- 0,010%
- 0,015%
- 0,020%
- 0,030%



AVALIAÇÃO DE DESEMPENHO

Resultados - 3 dias

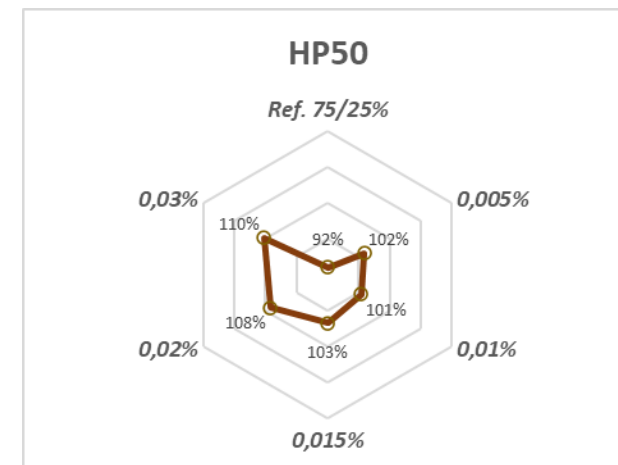
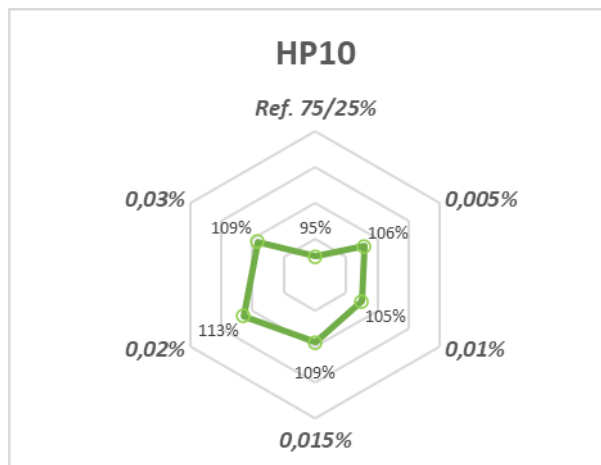
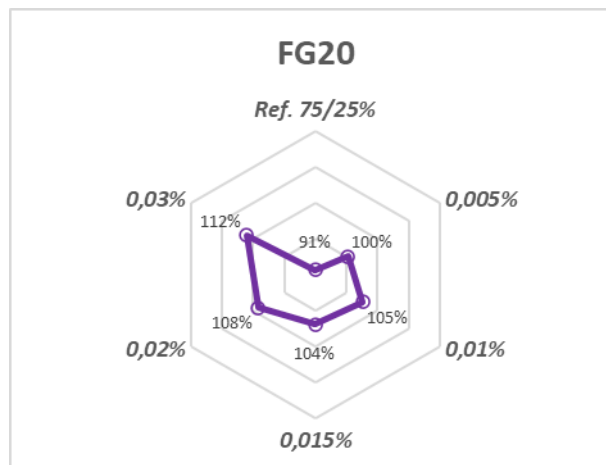
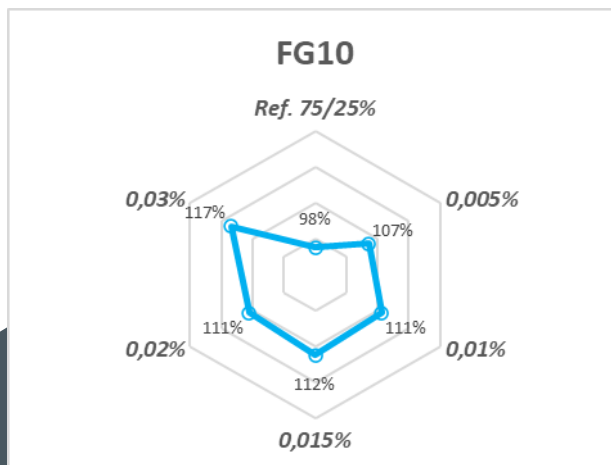
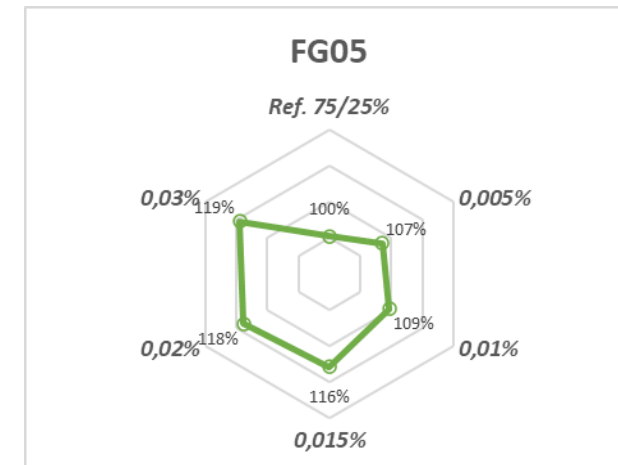
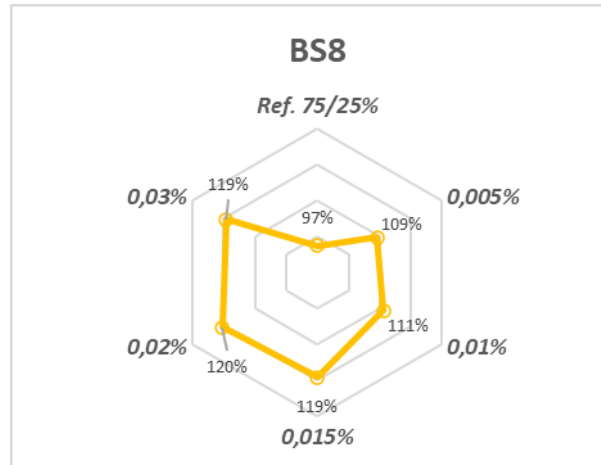
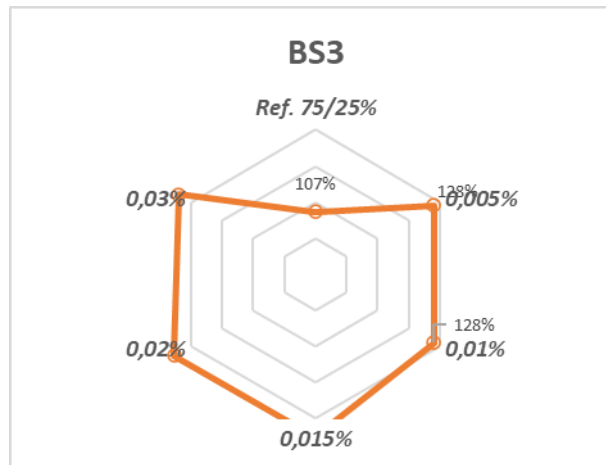
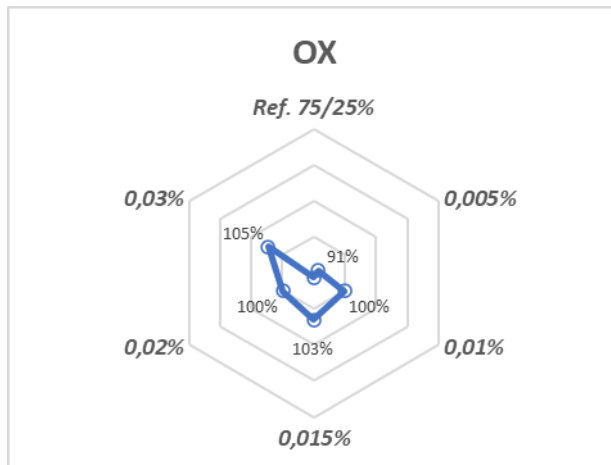
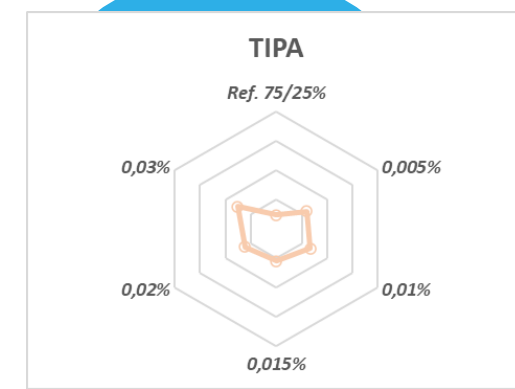
Percentual de resistência à compressão em relação a referência (100% cimento)



AVALIAÇÃO DE DESEMPENHO

Resultados - 7 dias

Percentual de resistência à compressão em relação a referência (100% cimento)

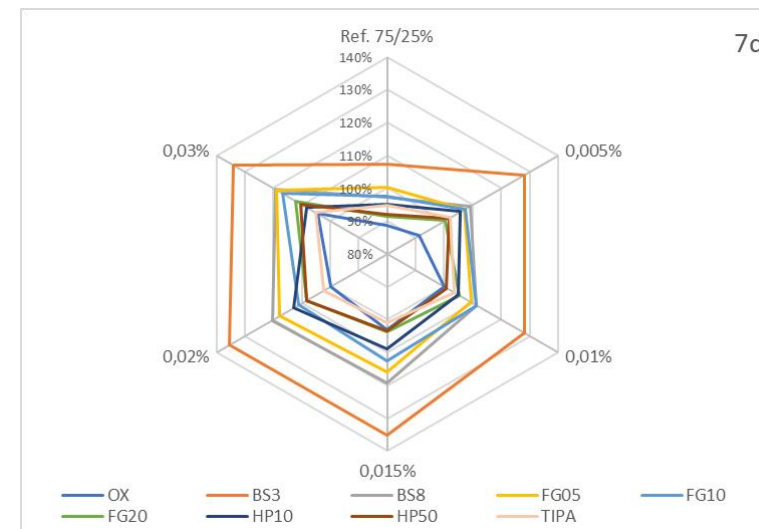
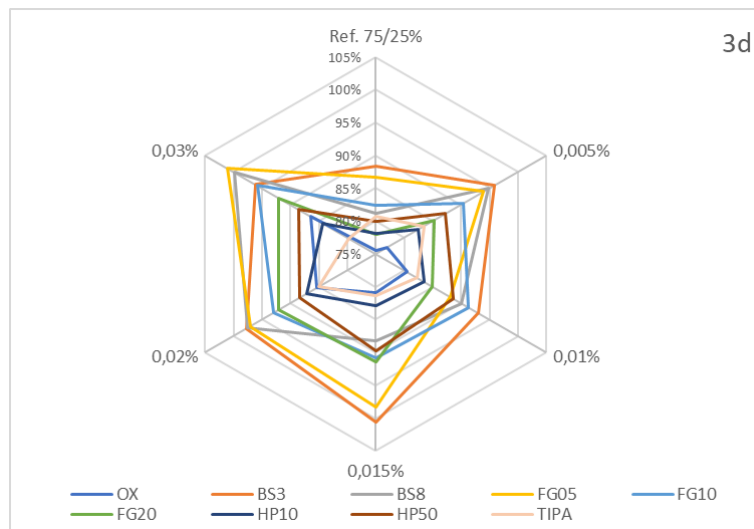
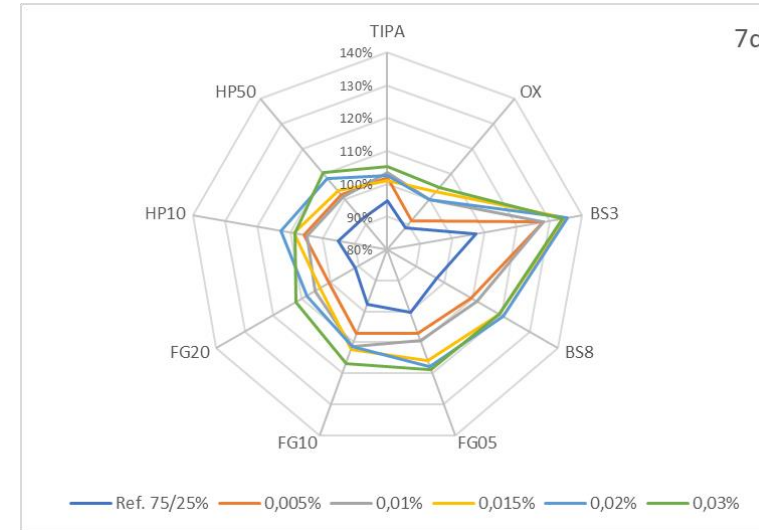
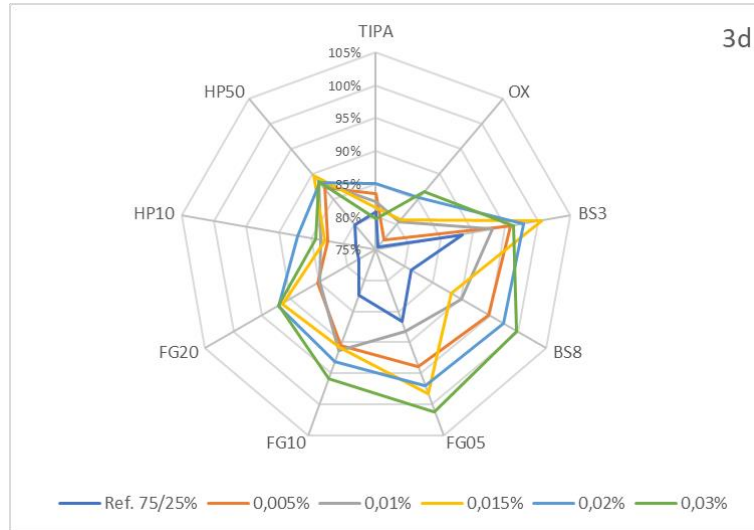


AVALIAÇÃO DE DESEMPENHO

Resultados



Percentual de resistência à compressão em relação a referência (100% cimento)





Obrigado!